INDIAN INSTITUTE OF ASTROPHYSICS

# Zero Shadow Day

*An activity book for schools and libraries*

IIA Outreach Team

2024 · English · CC BY-NC 4.0
Arivu Kendra sample edition · Government of Karnataka

# Contents

1. What is a zero shadow? 3
2. Dates for your district 8
3. Build a shadow stick 13
4. Measure and record 18
5. Upload your data 23
6. What the map shows 28

This is a generated sample edition created for the Arivu Kendra public preview. The text on these pages is placeholder prose so that reading tools, previews and downloads can be demonstrated end to end. The real edition will be supplied by the content provider.

CHAPTER 1

# What is a zero shadow?

Twice a year, for a few minutes around noon, your shadow disappears. This activity book explains why, tells you the exact dates for your district, and walks you through measuring shadow lengths with a stick, a tape and a phone so your class can upload its data to the state-wide map.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have

used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At

sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

CHAPTER 2

# Dates for your district

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your

phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays

together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright.

This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

CHAPTER 3

# Build a shadow stick

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have

used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At

sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

CHAPTER 4

# Measure and record

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your

phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays

together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright.

This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

CHAPTER 5

# Upload your data

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have

used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At

sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

CHAPTER 6

# What the map shows

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your

phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays

together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright.

This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.